

Rohit Kumar Sahoo

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Professional Summary

Computer Science undergraduate focused on real-time systems and applied AI, with hands-on experience building latency-sensitive audio and communication applications. Developed multiple production-style projects involving WebRTC, on-device ML, and event-driven architectures, with measured latency optimizations (<50ms processing in audio pipelines).

Technical Skills

Languages: Kotlin, Python, C++, Java

Frameworks: Jetpack Compose, Android SDK, TensorFlow Lite, ONNX Runtime, Node.js, Express.js

Systems: Real-Time Systems, Event-Driven Architecture, MVVM, REST APIs, WebRTC, JNI

Databases: Firebase Firestore, PostgreSQL, SQLite

Tools: Git, GitHub, Android Studio, Gradle, Firebase, Cloudinary

AI/ML: CNNs (YAMNet), Voice Activity Detection, Speaker Recognition, Audio Processing

Projects

Pausify – Real-Time Audio Control System [GitHub]

Dec 2025 – Feb 2026

Kotlin, C++, ONNX Runtime, TensorFlow Lite

- Reduced audio processing latency to <50ms on 16kHz streams by implementing a JNI-based C++ pipeline (baseline: ~120ms in pure Kotlin).
- Improved voice detection reliability in noisy environments by integrating VAD, YAMNet, and speaker recognition, reducing false triggers during testing scenarios.
- Designed a deterministic state machine to handle real-time audio events, ensuring consistent behavior under varying input conditions.

SoSafe – Real-Time Emergency Response System [GitHub]

Mar 2026 – Present

Kotlin, WebRTC, Firebase, Cloudinary

- Built a real-time SOS system enabling live audio streaming and location tracking using WebRTC and Firebase.
- Designed a hybrid streaming architecture (peer-to-peer + cloud fallback) to maintain connectivity under unstable network conditions.
- Achieved sub-second event synchronization across devices during real-time testing sessions.

AutoCommitBot – AI-Assisted Git Automation CLI [GitHub]

Feb 2026 – Mar 2026

Python, Typer, GitPython, Gemini API

- Developed a CLI tool to generate context-aware commit messages from code diffs using LLM APIs.
- Implemented fallback mechanisms to ensure consistent output during API failures.
- Automated commit workflows using Windows Task Scheduler, enabling zero-touch background logging.

Experience

AI/ML & App Development Lead – Enigma Coding Club

July 2025 – Present

- Coordinated development of student-led AI and Android projects, guiding implementation decisions and debugging workflows.

AI/ML Intern – CTTC Bhubaneswar

May 2025 – June 2025

- Built and evaluated machine learning models on structured datasets, including preprocessing and feature engineering.
- Assisted in model validation and performance analysis using standard evaluation metrics.

Education

B.Tech in Computer Science & Engineering

2024 – 2028

Veer Surendra Sai University of Technology, Odisha
CGPA: 8.38 / 10.0 (3rd Semester)